



PathLens: A lightweight multimodal reasoner for in-depth pathology insights

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ABSTRACT

Pathological diagnosis is crucial for clinical decision-making; however, most existing AI systems provide only direct predictions, lacking the detailed reasoning processes essential for clinical practice. To address this limitation, we introduce PathLens, a lightweight multimodal pathology reasoner designed to deliver interpretable and efficient diagnostic reasoning. Built upon HealthGPT, PathLens avoids costly pretraining by leveraging fine-tuning on synthetic data and emulates the step-by-step diagnostic workflow of pathologists in a transparent and clinically meaningful manner. This is achieved by Patho-CoT, a novel multi-agent pipeline that synthesizes high-fidelity multimodal Chain-of-Thought data, followed by Reasoning Alignment Learning, a training paradigm that integrates supervised and reinforcement learning to align the model with expert clinical workflows. To facilitate evaluation, we also introduce PedPathVQA, the first pilot dataset dedicated to pediatric pathology visual question answering (VQA). Experimental results demonstrate that PathLens achieves a favorable balance between performance and parameter efficiency, delivering competitive results across diverse pathology VQA benchmarks while maintaining substantially lower training costs than those of comparable or larger models. Additionally, PathLens offers a clinically deployable alternative that avoids patient data transmission to external servers, addressing a key requirement in privacy-sensitive clinical environments. The code is available at <https://github.com/hovchen/PathLens>.

1. Introduction

Pathological diagnosis is a fundamental pillar of clinical medicine, playing a critical role in disease classification, survival prediction, and prognosis assessment [1–4]. In routine practice, these diagnostic decisions often rely on the careful examination of subtle morphological patterns that can vary significantly across cases and disease stages [5]. Pathologists must interpret large volumes of heterogeneous visual information, in which both fine-grained cellular features and broader tissue architecture are essential for accurate assessment [6]. With the rise of digital pathology and deep learning, substantial progress has been made in automating analytical tasks such as cancer grading [7] and cell segmentation [8]. However, these systems typically produce one-off predictions, whereas Visual Question Answering (VQA) introduces a question-driven interaction paradigm, opening a pathway

toward more flexible diagnostic reasoning. In this context, pathology VQA has emerged as a novel and promising research direction [9]. Unlike conventional classification or segmentation pipelines, VQA allows pathologists to actively query patch-level pathology images with flexible and clinically relevant questions. For example, a VQA system may be asked not only to determine whether a patch contains malignant cells, but also to describe the morphological cues supporting that decision. This capability brings AI systems closer to mimicking expert reasoning, as it requires both precise visual grounding and the articulation of diagnostic logic. Ultimately, such systems could augment pathologists' decision-making processes, reduce cognitive workload, and enhance trust by providing transparent explanations rather than black-box outputs [10].

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To support these capabilities, vision-language pretraining approaches have laid an important foundation. Early CLIP-like [11] models demonstrated that large-scale alignment between images and text can yield powerful representations for pathology, enabling tasks such as retrieval and zero-shot classification [12]. Building on this, Multimodal Large Language Models (MLLMs) have further advanced the paradigm by tightly integrating visual and linguistic reasoning, offering unprecedented flexibility for multimodal tasks [13–15]. However, translating these models into real-world clinical applications reveals two major challenges. First, their reasoning processes often remain opaque, undermining transparency and limiting clinical trust [16–19]. Second, the substantial computational cost of training and inference makes it difficult to deploy such models in resource-constrained healthcare environments, where efficiency and sustainability are essential [20]. These limitations indicate that progress in pathology VQA cannot rely solely on brute-force scaling, but instead requires domain-specialized, efficient, and interpretable approaches to meet the practical demands of clinical pathology.

Building on this motivation, we introduce PathLens, a lightweight yet effective multimodal pathological reasoner designed to deliver domain-specific reasoning with minimal computational overhead. Rather than being trained from scratch, PathLens leverages HealthGPT [21], a publicly available medical MLLM, thereby inheriting a broad foundation of general medical knowledge. This strategy avoids redundant relearning of general medical priors and instead focuses model capacity on pathology-specific reasoning. To enable this specialization, we first develop Patho-CoT, an automated multi-agent pipeline that synthesizes high-quality multimodal Chain-of-Thought (CoT) traces [22–25]. Each trace explicitly couples patch-level visual observations with concise clinical rationales, allowing the model to emulate how human pathologists articulate their diagnostic logic. The resulting dataset forms the basis of our *Reasoning Alignment Learning* (ReAL) strategy. ReAL integrates supervised fine-tuning (SFT) [26] and subsequent Reinforcement Learning with Verifiable Reward (RLVR) [27], thereby enabling PathLens to emulate the diagnostic process of a pathologist in a step-by-step and interpretable manner. To support rigorous and clinically meaningful evaluation, we also construct PedPathVQA, which, to the best of our knowledge, is the first pilot dataset dedicated to pediatric pathology VQA. This dataset addresses a long-standing gap in publicly available resources for this important yet underserved domain.

The primary contributions of this work are as follows:

- We present **Patho-CoT**, a multiagent collaborative pipeline that automatically synthesizes high-quality multimodal CoT data for pathology, enabling explicit and interpretable reasoning.
- We propose **PathLens**, a lightweight pathological reasoner that achieves strong performance through the **ReAL** strategy. By fine-tuning a medical MLLM with synthesized CoT data, it avoids costly pretraining while ensuring stepwise interpretability.
- We extensively validate PathLens on multiple pathology VQA benchmarks, along with the introduction of **PedPathVQA**, the first pilot benchmark for pediatric pathology VQA, to further assess model generalization.

2. Related work

2.1. Pathology VQA

Pathology VQA has recently emerged as a key direction in medical AI, aiming to automatically answer natural language questions based on pathological images. The earliest effort in this domain was the PathVQA dataset proposed by He et al. [9], which established the first benchmark with 4998 images and 32,799 QA pairs. Building on this foundation, subsequent work [28] progressed from dataset construction to model training, introducing a three-stage framework with a “learning-by-ignoring” strategy that effectively mitigates the impact

of noisy annotations. To further enhance medical reasoning, Naseem et al. [29] proposed K-PathVQA, which explicitly incorporates medical knowledge graphs to integrate visual and semantic information, significantly improving open-ended answering accuracy. Recognizing the need for multimodal and multilingual communication in real clinical settings, Rajkhowa et al. [30] extended PathVQA to TM-PathVQA, enabling speech-based interaction across languages. In parallel, WSI-VQA [31] expanded the scope from patch-level analysis to whole-slide images (WSIs), creating a new benchmark for WSI-level reasoning. To better align with clinical teaching scenarios, Seyfioglu et al. [32] developed Quilt-VQA from instructional pathology videos through automatic extraction and manual validation of QA pairs. More recently, Sun et al. [33] released PathMMU, an expert-validated benchmark comprising 33,428 multiple-choice questions (MCQs) across 24,067 images, further standardizing evaluation across tasks. The steady evolution of these benchmarks has not only expanded dataset scale and diversity but also exposed the limitations of task-specific models, thereby motivating the development for foundation-level approaches capable of capturing richer multimodal semantics.

2.2. Multimodal large models in pathology

With the advent of multimodal large models (MLMs), pathological image understanding has entered a new era, characterized by large-scale pretraining and instruction tuning. Early explorations primarily followed a CLIP-like discriminative paradigm. For instance, Lu et al. [34] introduced MI-Zero, a multiple-instance learning (MIL) framework that partitions WSIs into bags of instances, enabling zero-shot cancer subtyping. Building upon this direction, Huang et al. [12] proposed PLIP, trained via contrastive learning on more than 200,000 pathology image-text pairs collected from Twitter, demonstrating strong zero-shot retrieval and classification performance. Extending this line of work, Lu et al. [35] presented CONCH, which combines contrastive and generative objectives on 1.17 million image-caption pairs, thereby improving generalization across diverse cancer subtypes. Around the same time, Wang et al. [36] developed CHIEF, a foundation model trained on more than 60,000 WSIs, shifting the focus from patch-level learning to WSI-level representation and enabling downstream tasks such as origin identification, genomic profiling, and prognosis prediction. To further enhance model versatility, Xiang et al. [37] introduced MUSK, a vision-language foundation model that performs strongly not only in retrieval and VQA but also in biomarker prediction and clinical outcome forecasting. Despite these advances, CLIP-like architectures remain inherently limited by their lack of generative reasoning capability, constraining their applicability in interactive tasks such as VQA. To overcome this limitation, recent work has incorporated LLMs into pathology and related domains [3,38,39]. Seyfioglu et al. [32] proposed Quilt-LLaVA, which aligns narrated pathology videos with 107K instruction pairs to enable spatially grounded reasoning across WSI patches, surpassing prior VQA systems. Building on this shift, Lu et al. [40] developed PathChat, a pathology copilot that integrates a large vision encoder with Llama-2 and is fine-tuned on 456K instruction samples, achieving state-of-the-art performance in diagnostic and open-ended tasks. Similarly, Dai et al. [41] advanced the paradigm with PA-LLaVA, replacing CLIP with a PLIP encoder and designing a scale-invariant connector, thereby improving both supervised and zero-shot VQA performance. Moving toward generalist assistants, Xiang et al. [42] presented CPath-Omni, which unifies diagnosis, prognosis, and biomarker prediction across multimodal inputs. Most recently, Sun et al. [43] introduced PathGen-LLaVA, trained on the synthetic PathGen-1.6M dataset generated through multiagent collaboration. While these developments significantly enhance the capability and interactivity of pathology models, they also reveal a critical limitation: the lack of structured and interpretable reasoning in clinical contexts. Addressing this gap has therefore become a key motivation for the growing adoption of CoT methods.

2.3. CoT reasoning in pathology

CoT reasoning has recently attracted significant attention as it enhances both the interpretability and reliability of advanced AI systems, particularly in safety-critical domains. Early studies such as DeepSeek-Math [27] and DeepSeek-R1 [44] demonstrated that RLVR can substantially improve CoT reasoning in LLMs, inspiring the medical AI community to explore similar strategies. For example, Zhang et al. [45] proposed Med-U1, which employed RLVR with multi-objective rewards to strengthen medical reasoning across diverse task formats. Lai et al. [46] extended this paradigm to multimodal domains through Med-R1, applying Group Relative Policy Optimization (GRPO) with rule-based rewards to elicit structured reasoning. Similarly, Pan et al. [47] introduced MedVLM-R1, achieving competitive performance while requiring substantially less training data, and Xu et al. [48] developed Lingshu, which incorporated both curated and synthetic datasets within a multi-stage pipeline to align reasoning with medical domain knowledge. These works collectively highlight the central role of CoT in medical AI, although they primarily focus on general medical tasks. More recently, pathology has emerged as a critical domain for its application, motivating efforts to adapt and refine CoT-based approaches for pathological analysis. For instance, Wu et al. [49] proposed PathVLM-R1, which integrates GRPO with both outcome-level and process-level rewards to elicit structured, pathology-aware CoT. Building on this work, Zhang et al. [50] introduced Patho-R1, advancing the paradigm further by integrating through a three-stage pipeline, that enables more reliable pathology-specific reasoning across diverse tasks. Taken together, these studies illustrate that CoT reasoning, empowered by RLVR, is increasingly becoming a cornerstone for pathology-oriented multimodal intelligence, laying the foundation for the development of trustworthy clinical AI.

3. Methods

3.1. Preliminaries: The HealthGPT model

HealthGPT [21] is a medical MLLM designed to unify visual comprehension and visual generation within a single autoregressive framework. A key innovation is in the Heterogeneous Low-Rank Adaptation (H-LoRA) mechanism, which introduces task-specific LoRA modules to decouple comprehension and generation knowledge. This design mitigates interference between heterogeneous tasks and enables efficient adaptation across diverse medical domains.

For visual encoding, HealthGPT employs a CLIP-L/14 backbone, extracting both shallow and deep features through lightweight MLP adapters to achieve effective vision–language alignment. To further enhance multimodal coordination, HealthGPT uses a three-stage learning strategy that progressively aligns comprehension and generation modules in a decoupled yet unified manner, improving both stability and generalization across tasks.

3.2. Overview of PathLens

While HealthGPT provides a strong foundation for multimodal medical understanding, its reasoning capability in fine-grained pathology remains limited. To address this gap, we introduce **PathLens**, a model that substantially enhances visual reasoning in digital pathology via our proposed **ReAL** strategy. As illustrated in Fig. 1, the overall framework operates in two distinct phases: data preparation via Patho-CoT and model training via ReAL.

We first employ the Patho-CoT pipeline to curate data from multiple sources and apply tailored processing strategies, constructing a high-quality corpus that includes both VQA and CoT supervision data. The model then undergoes a two-stage ReAL training procedure. In ReAL Stage I, HealthGPT is fine-tuned with H-LoRA to teach the model to “reason-before-answering”. In ReAL Stage II, its reasoning capability is

further strengthened via a carefully designed reward mechanism using Group Sequence Policy Optimization (GSPO) [51].

In contrast to prior methods that focus primarily on either visual perception or language modeling in isolation, PathLens emphasizes end-to-end interpretability and clinical reliability. By integrating structured CoT reasoning directly into the model’s generation process, PathLens not only improves answer accuracy but also provides transparent decision pathways that align with real-world diagnostic reasoning workflows. Moreover, RL-based refinement ensures that the model continues to enhance its reasoning fidelity beyond supervised signals, preserving generalization ability across diverse pathology scenarios. Overall, PathLens offers a scalable and domain-adaptive framework that bridges the gap between MLLMs and clinically trustworthy computational pathology assistants.

3.3. Patho-CoT: Multi-Agent collaboration for pathology CoT construction

As illustrated in Fig. 2, Patho-CoT is instantiated as a multiagent collaboration framework, in which a set of specialized agents with complementary capabilities cooperatively construct clinically grounded CoT reasoning. The overall pipeline consists of two data sourcing strategies, followed by a multiagent synthesis process. All utilized data are strictly disjoint from the evaluation set to ensure fair and reliable performance assessment.

Data sourcing agents. To bootstrap the pathology VQA corpus, we employ two complementary sourcing strategies. The *Filtering Agent* applies a lightweight pathology image classifier [52] to existing medical VQA datasets (e.g., PMC-VQA [53], PathVQA [9]), extracting relevant pathology subsets. In parallel, the *Generation Agent* produces VQA samples from pathology instruction-tuning datasets (e.g., PathInstruct [52]) using Qwen-VL [54], with validation by board-certified pathologists. Together, these agents collaboratively curate a high-quality pathology VQA dataset. From this pool, we select 10,000 samples for CoT construction, reserving the remainder for subsequent RLVR.

Visual feature agent. Powered by DeepSeek-R1 [44], this agent operates on QA pairs to infer the essential visual cues underpinning clinical reasoning. While it lacks direct image perception, its strong semantic reasoning capacity allows it to hypothesize critical visual regions or structures, producing key features that subsequently serve as semantic priors for downstream agents.

Caption agent. Implemented with HealthGPT model, this agent leverages both the pathology image and the semantic priors to generate enhanced image captions. By jointly modeling visual and textual context, it emphasizes clinically significant regions and adheres to standard medical terminology. The resulting descriptions provide a comprehensive characterization of anatomical structures, suspicious lesions, and their spatial distribution, bridging semantic hypotheses with visual evidence.

Reasoning agent. Re-engaging DeepSeek-R1, this agent is now conditioned on the enhanced description, the question, and the reference answer. It generates reasoning chains in the format `<think> . . . </think><answer> . . . </answer>`, where the `<think>` encodes step-by-step diagnostic reasoning and `<answer>` provides the final conclusion. This process integrates descriptive evidence with clinical inference, ensuring interpretability and traceability.

Refinement agent. To address potential artifacts in reasoning traces (e.g., explicit mentions of “image description” or “standard answer”), the Refinement Agent employs DeepSeek-V3 to post-process outputs. It removes artificial cues, enforces direct grounding in visual content, and polishes the narrative flow. This self-critique mechanism enhances both the plausibility and realism of the reasoning chains.

To ensure the clinical reliability and factual correctness of the synthesized CoT reasoning traces, we also incorporate an expert validation

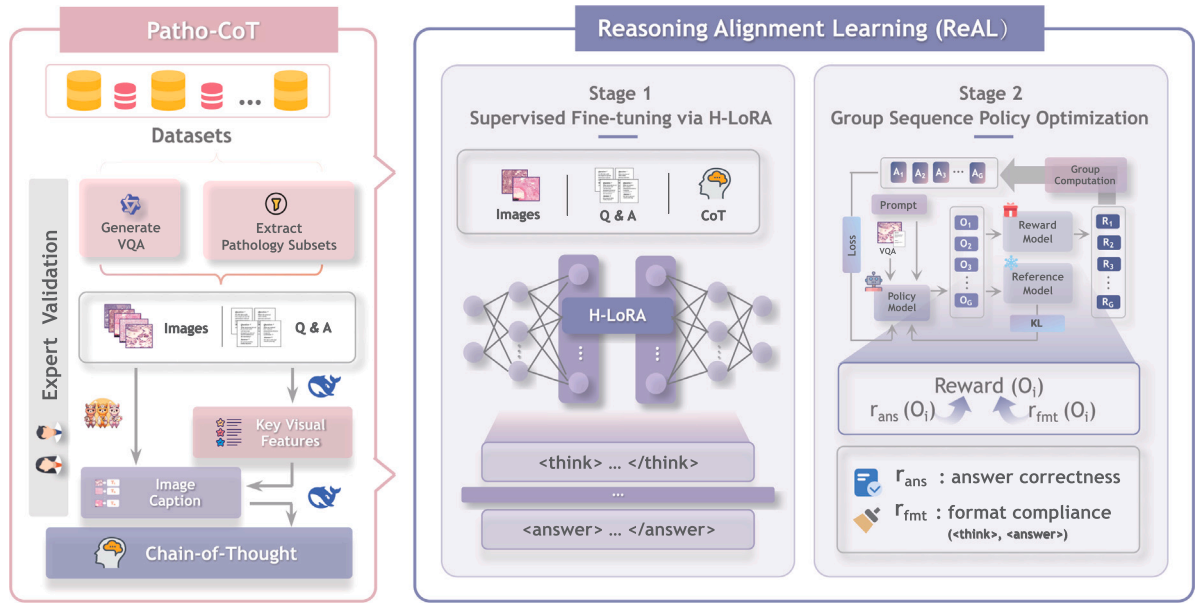


Fig. 1. The overall framework of PathLens comprises two main phases: data construction and model training. The left panel displays Patho-CoT, which constructs high-quality CoT data. This module processes pathology datasets through expert-validated VQA generation and subset extraction, utilizing multi-modal models to synthesize image descriptions and key visual features into detailed reasoning paths. The right panel illustrates the Reasoning Alignment Learning (ReAL) strategy. Stage 1 involves SFT via H-LoRA, training the model on images, Q&A pairs, and the generated CoT to produce structured outputs containing distinct thinking and answer sections. Stage 2 implements GSPO, which utilizes a reward mechanism that evaluates group outputs based on answer correctness (r_{ans}) and format compliance (r_{fmt}) to align the reasoning process with clinical standards.

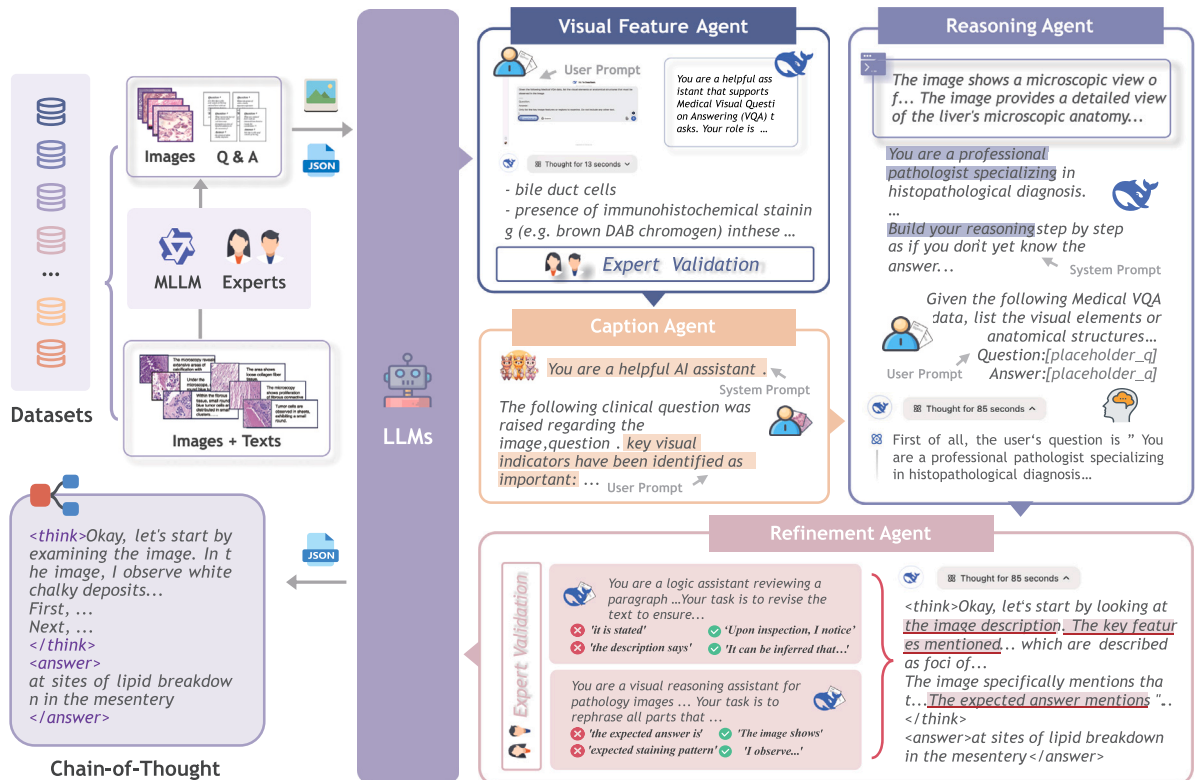


Fig. 2. Overview of the multi-agent Patho-CoT pipeline. The framework consists of two main phases: Data Sourcing (left) and CoT Synthesis (right). In the synthesis phase, four specialized agents collaborate: (1) the Visual Features Agent identifies anatomical structures; (2) the Caption Agent provides contextual descriptions; (3) the Reasoning Agent simulates a professional pathologist's diagnostic process; and (4) the Refinement Agent polishes the text to ensure a logical, observational tone. The final output is a structured CoT sequence.

mechanism within the Patho-CoT pipeline. Specifically, board-certified pathologists are involved at critical stages of data construction to review a subset of generated samples, verifying their medical accuracy and consistency with real-world diagnostic workflows.

In summary, the Patho-CoT pipeline embodies multi-agent collaboration, where agents specializing in feature hypothesis, medical description, reasoning construction, and consistency refinement operate in synergy. Their coordinated interaction produces clinically interpretable, trustworthy, and high-quality CoT annotations for pathology VQA tasks.

3.4. ReAL

To equip PathLens with interpretable and clinically faithful reasoning capabilities, we propose the ReAL strategy. This paradigm unifies supervised initialization and RL refinement into a cohesive two-stage process.

Stage I: SFT via H-LoRA

We initiate the strategy by adapting HealthGPT to a “reason-before-answering” paradigm, which encourages explicit intermediate reasoning prior to prediction. During this stage, high-quality CoT supervision synthesized by Patho-CoT is employed to guide the model toward structured, pathology-consistent diagnostic reasoning. Rather than learning only surface-level output patterns, the model is trained to follow step-by-step diagnostic logic that reflects lesion assessment, evidence integration, and conclusion formulation in a coherent and clinically informed manner.

In this configuration, low-rank adapters are injected into the backbone while the vision tower remains frozen, enabling parameter-efficient adaptation without disrupting pretrained visual representations. To explicitly disentangle the reasoning process from the final answer, we introduce four special tokens: $\langle \text{think} \rangle$, $\langle / \text{think} \rangle$, $\langle \text{answer} \rangle$, and $\langle / \text{answer} \rangle$. The embeddings of these tokens are initialized as the average of the representations of the existing “ $\langle$ ” and “ $\rangle$ ” symbols to promote optimization stability. Additionally, elevated loss weights are applied to these boundary tokens to reinforce structural consistency and maintain a clear separation between reasoning and answer segments.

The model is trained on VQA instances (I, q) with the supervised target formatted as follows:

$$y = (\langle \text{think} \rangle, r_{1:T_r}, \langle / \text{think} \rangle, \langle \text{answer} \rangle, a_{1:T_a}, \langle / \text{answer} \rangle), \quad (1)$$

where $r_{1:T_r}$ denotes the reasoning trace and $a_{1:T_a}$ the final answer. Using a weighted cross-entropy objective, this stage establishes a structured reasoning prior and provides a strong initialization for the subsequent reinforcement learning stage.

Stage II: GSPO

Following SFT, we refine the model using GSPO to optimize reasoning quality at the sequence level. Unlike token-wise methods (e.g., PPO, GRPO [27]), GSPO aligns optimization granularity with the unit of reward, thereby improving stability. Given an input (I, q) , the model generates G candidate responses $\{O_i\}_{i=1}^G$. We design a dedicated reward function based on VQA characteristics:

$$\text{Reward}(O_i) = r_{\text{ans}}(O_i) + r_{\text{fmt}}(O_i), \quad (2)$$

where r_{ans} rewards correct predictions, and r_{fmt} enforces the required format.

Within each group, the advantage $\hat{A}_i$ is computed via normalization. GSPO then updates the model by maximizing the advantage-weighted and clipped sequence likelihood ratio:

$$J_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_i), \quad (3)$$

where $s_i(\theta)$ is the length-normalized likelihood ratio between the current and old policies. This stage is crucial for suppressing hallucinations and ensuring that the generated reasoning is not only structurally correct but also clinically grounded.

4. Experiments

4.1. Datasets

We evaluate our model on multiple pathology VQA benchmarks to assess both in-domain performance and out-of-distribution generalization. Below, we summarize the datasets used in our experiments.

PathVQA. PathVQA [9] is a widely used pathology VQA benchmark comprising 4998 pathology images paired with 32,799 manually verified QA pairs spanning both open-ended and multiple-choice formats. The questions cover basic visual recognition, descriptive interpretation, and clinically oriented reasoning. Following prior work, we adopt the standard dataset split and report performance on the test set.

PathMMU. PathMMU [33] is a large-scale, expert-curated multimodal pathology benchmark designed to evaluate clinical reasoning. The dataset contains 24,067 images and 33,428 multiple-choice questions, each accompanied by professionally reviewed answer explanations. The dataset aggregates diverse, authoritative sources, including pathology atlases, textbooks, research publications, educational materials, and social media case discussions. All QA pairs are rigorously validated by board-certified pathologists to ensure diagnostic accuracy. We report performance on both the validation and test splits following the official protocol.

Quilt-VQA. Quilt-VQA [32] is derived from narrated histopathology teaching videos, where key visual regions are aligned with expert instructional narrations. The dataset contains 985 images with 1283 high-quality QA pairs emphasizing morphological feature identification and slide-level diagnostic reasoning. Compared to datasets derived solely from static captions, Quilt-VQA reflects more authentic clinical teaching scenarios and reasoning processes. We evaluate directly on the officially provided benchmark without additional subset division.

PedPathVQA (Ours). To assess model generalization on pediatric pathology, we construct PedPathVQA, the first pilot dataset dedicated to pediatric pathology VQA. As illustrated in Fig. 3, the H&E staining WSIs are digitized at high resolution using whole-slide scanners. To ensure clinical fidelity and diversity, licensed pediatric pathologists review the slides, manually select diagnostically meaningful regions, and compose clinically relevant QA pairs, which constitute the core annotations of our VQA dataset. Following the initial annotation, a committee of senior pathologists performs cross-validation to ensure questions are clinically meaningful and answers are strictly evidence-based. Ultimately, this rigorous curation results in the final PedPathVQA benchmark comprising 541 high-quality VQA instances covering 54 distinct pediatric disease entities.

To address the challenge of cross-domain generalization with limited data volume, we have designed PedPathVQA to maximize diversity and granularity. The dataset spans 10 major organ systems (including the nervous, urinary, digestive, blood/lymph & immune, skin & appendages, soft tissue, muscle & bone, non-specific pathological changes & normal controls, reproductive system & embryonic tumors, respiratory, and endocrine systems), effectively establishing a cross-domain testbed. Fig. 4 shows the detailed statistical distribution. It can be seen that, beyond malignant tumors, inflammatory conditions, immune diseases, congenital and developmental abnormalities, as well as benign lesions with reactive hyperplasia and normal tissue are also included to enhance the diversity within the domain. Moreover, within the same disease, we also incorporate image examples encompassing different pathological subtypes and control conditions of healthy patients. This diversified design at different granularities ensures that models are

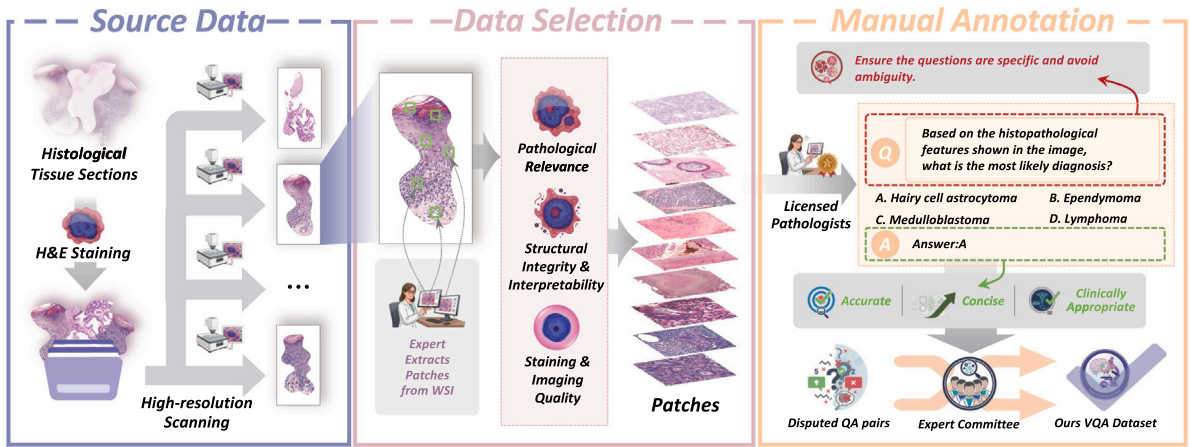


Fig. 3. PedPathVQA tissue section processing workflow. Formalin-Fixed, Paraffin-Embedded (FFPE) sections are H&E-stained and digitized into WSIs. Pathologists select diagnostically meaningful tissue regions from the screened WSIs and crop them into patches. Licensed pathologists then generate QA pairs based on directly observable histological features, followed by cross-review by a senior expert committee to ensure annotation consistency and clinical reliability.

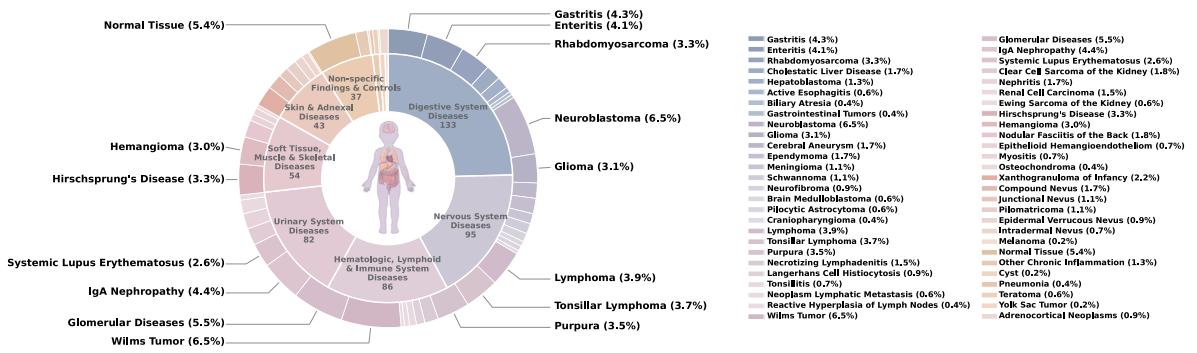


Fig. 4. Hierarchical distribution of the PedPathVQA dataset across different disease categories. The inner ring summarizes high-level organ-system groups (e.g., digestive, nervous, urinary systems), with the corresponding number of samples annotated for each category. The outer ring further breaks down these groups into fine-grained disease subtypes, illustrating the dataset's detailed composition and class imbalance.

evaluated on their ability to distinguish different pathological lesion types and healthy histological features. Only those possess the capability to discern subtle pathological features rather than relying solely on high-level context at the organ level can perform well.

4.2. Implementation details

We implement PathLens using PyTorch and conduct training on 4×48 GB vGPUs. The vision encoder remains frozen throughout training, while LoRA adapters are inserted into the language backbone with a rank of 64 and $\alpha = 128$ to enable parameter-efficient fine-tuning.

For memory-efficient distributed training, we employ DeepSpeed ZeRO-2. The SFT stage is optimized using AdamW with a learning rate of 2×10^{-5} , weight decay of 0.01, and cosine learning rate decay. The SFT stage is trained for 3 epochs with a global batch size of 32, followed by a learning rate refinement stage for 1 epoch at a reduced learning rate of 5×10^{-7} . Gradient clipping is applied with a maximum norm of 1.0, and BF16 mixed-precision training is enabled to reduce memory consumption.

During the RL stage, we adopt the ms-swift framework on top of our DeepSpeed setup. All experiments are conducted under consistent configurations across datasets to ensure comparability. The detailed hyperparameter configuration of the RL stage is provided in Appendix D.

4.3. In-domain performance

We first evaluate the in-domain performance of PathLens on three representative pathology VQA benchmarks: PathVQA, PathMMU, and

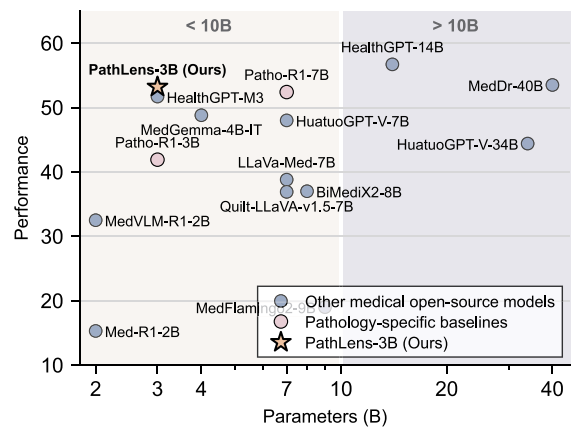


Fig. 5. Performance comparison on PathVQA across medical MLLMs with varying parameter scales. PathLens achieves strong performance despite its small size, highlighting an efficient performance-parameter trade-off compared to both smaller and larger models.

Quilt-VQA. Together, these datasets cover basic visual recognition, fine-grained morphological reasoning, and clinically-oriented diagnostic interpretation, providing a comprehensive view of the model's pathology understanding and reasoning capabilities.

Table 1

Comparison of PathLens with state-of-the-art MLLMs on the PathVQA benchmark. Models are grouped into three categories: *Proprietary*, *Open-source General*, and *Open-source Medical*. We report answer accuracy (Acc) and parameter counts when available.

Type	Model	Acc	Model	Acc	Model	Acc	Params
Proprietary	GPT-4.1 [58]	55.5	Claude Sonnet [59]	54.2	Gemini-2.5 [60]	55.4	–
Open-source general	Qwen2.5VL-7B [54]	44.1	InternVL2.5-8B [61]	42.1	InternVL3-8B [62]	48.6	<10B
	Qwen2.5VL-32B [54]	41.9	InternVL2.5-38B [61]	46.9	InternVL3-38B [62]	51.0	>10B
Open-source medical	Med-R1-2B [46]	15.3	MedVLM-R1-2B [63]	32.5	MedGemma-4B-IT [64]	48.8	<10B
	HealthGPT-M3 [21]	51.7	LLaVa-Med-7B [65]	38.8	HuatuoGPT-V-7B [66]	48.0	
	Quilt-LLaVA-v1.5-7B [32]	36.9	BiMediX2-8B [67]	37.0	MedFlamingo2-9B [68]	19.0	
	Patho-R1-3B [50]	41.9	Patho-R1-7B [50]	52.4	PathLens-3B (Ours)	53.2	>10B
	HealthGPT-14B [21]	56.7	HuatuoGPT-V-34B [66]	44.4	MedDr-40B [69]	53.5	

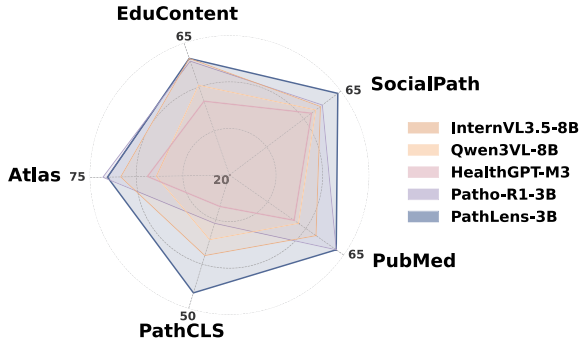


Fig. 6. Radar-chart comparison of PathLens with Patho-R1-3B, HealthGPT-M3, Qwen3-VL-8B, and InternVL3.5-8B on the PathMMU validation split. PathLens demonstrates consistently higher accuracy than other 3B-scale pathology or medical models, and achieves competitive performance against larger general-purpose MLLMs across most subsets.

PathVQA. As reported in Table 1, PathLens achieves an Acc of 53.2% on PathVQA. As illustrated in Fig. 5, PathLens demonstrates a favorable performance-parameter trade-off, achieving highly competitive results despite its small model size. Among open-source medical models with fewer than 10 billion parameters, this result surpasses MedFlamingo2-9B (19.0%), HuatuoGPT-V-7B (48.0%), BiMediX2-8B (37.0%), and LLaVa-Med-7B (38.8%). It also outperforms the pathology reasoning model Patho-R1-3B (41.9%), which operates at a comparable scale. Furthermore, when evaluated against larger-scale systems in the open-source landscape, such as HealthGPT-14B (56.7%) and Patho-R1-7B (52.4%), PathLens remains highly competitive. Despite having fewer than half as many parameters as Patho-R1-7B, PathLens delivers comparable results. This outcome indicates that domain-aligned CoT supervision can partially offset the advantages typically associated with increased model capacity. Finally, compared with proprietary models, a performance gap of approximately 2 percentage points remains relative to GPT-4.1 (55.5%) and Gemini-2.5 (55.4%). However, it is important to note that these proprietary systems require transmitting patient histological images to external cloud servers. This requirement raises significant privacy and data governance concerns in clinical settings, potentially precluding the adoption of such models in many hospital environments under regulations such as HIPAA [55] and GDPR [56]. By contrast, PathLens is fully deployable locally at modest hardware cost, providing a privacy-preserving alternative suitable for resource-constrained institutions [20,57].

PathMMU. We further evaluate the performance of PathLens on the PathMMU benchmark. As demonstrated in Table 2, the proposed model achieves competitive results across both validation and test sets when compared to several strong open-source MLLMs. On the validation split, PathLens attains an accuracy of 61.3%, outperforming medical and pathology models such as HealthGPT-M3 (49.4%) and Patho-R1-3B (58.4%). On the test split, the model maintains robust performance, reaching an accuracy of 64.4% on the Tiny subset and 61.1% on the full test set. While PathLens demonstrates an overall advantage over models

of a comparable scale, Patho-R1-7B achieves a higher overall validation accuracy (63.0%) and surpasses PathLens on specific test subsets, most prominently on the Atlas subset (75.3% compared to 61.5%). The Atlas subset primarily comprises high-resolution, expert-annotated morphological reference images, heavily emphasizing encyclopedic knowledge retrieval. Since the training paradigm of PathLens prioritizes complex clinical reasoning over the memorization of static morphological features, models with larger parameter capacities inevitably hold an advantage in such retrieval-intensive tasks. This performance variance is discussed comprehensively in Section 5.4. Despite this deviation on specific reference-heavy subsets, Fig. 6 visualizes the performance across various validation subsets of PathMMU, demonstrating that PathLens consistently maintains an advantage over pathology and medical models of a comparable scale, as well as over larger general-purpose models, across the majority of these subsets. These results indicate that the combination of synthetic pathology-specific CoT supervision with RL refinement serves as a parameter-efficient strategy for the enhancement of clinically aligned reasoning, even though the absolute performance relative to larger models exhibits variance across individual subset types.

Quilt-VQA. For this dataset, we follow the evaluation protocol of Zhang et al. [50], employing an LLM-as-a-judge framework together with fine-grained reasoning metrics. These include commonsense rationality, faithfulness, hallucination rate, information sufficiency at the step-level, missing-step detection, reasoning alignment, redundancy, repetition, and semantic coverage. As shown in Table 3, PathLens-3B achieves the strongest overall performance among all compared models, including both general-purpose MLLMs and domain-specific pathology models. In particular, it substantially surpasses previous open-source pathology models, such as Patho-R1-3B and Patho-R1-7B, especially on metrics closely associated with transparent clinical reasoning. The model also obtains competitive redundancy and repetition scores, demonstrating that its longer reasoning chains remain concise, focused and informative. These consistent gains across nearly all dimensions show that PathLens delivers more structured, faithful, and clinically aligned visual reasoning on challenging pathology VQA tasks compared to existing solutions.

4.4. Cross-domain generalization

Pediatric pathology exhibits distinct morphological characteristics and diagnostic patterns compared to the datasets commonly used in pathology VQA research. To assess whether PathLens can generalize beyond its training distribution, we evaluate it on PedPathVQA, the first pilot dataset dedicated to pediatric histopathology VQA.

Unlike the datasets used during training, PedPathVQA contains tissue structures, lesion types, and clinical contexts specific to pediatric cases. This domain shift introduces novel visual appearances and reasoning requirements not directly reflected in the training data. Evaluating on PedPathVQA therefore provides a direct measure of cross-domain robustness.

As shown in Table 4, PathLens-3B achieves a leading accuracy of 60.07% among all open-source models. Remarkably, despite having significantly fewer parameters (3B), PathLens outperforms larger

Table 2

Comparison of **PathLens** with other MLLMs on PathMMU benchmark. Models are grouped into three categories: *Proprietary*, *Open-source General*, and *Open-source Medical*. We report the answer Acc in this table.

Model	Val (710)	Test		PubMed		SocialPath		EduContent		Atlas		PathCLS	
		Tiny (1156)	ALL (9677)	Tiny (281)	ALL (3068)	Tiny (235)	ALL (1855)	Tiny (255)	ALL (1938)	Tiny (208)	ALL (1007)	Tiny (177)	ALL (1809)
Expert	–	71.8	–	72.9	–	71.5	–	69.0	–	68.3	–	78.9	–
Proprietary													
Gemini Pro Vision [60]	41.9	42.8	42.7	43.8	44.9	42.4	42.0	43.5	43.7	49.5	49.4	32.8	34.7
GPT-4V [58]	49.3	53.9	49.8	59.4	53.5	58.7	53.9	60.4	53.6	48.1	52.8	36.2	33.8
GPT-4 Turbo [58]	36.5	41.8	38.1	48.8	42.4	43.4	42.3	47.1	40.6	41.3	43.2	22.3	21.1
Open-source general													
LLaVA-1.5-7B [70]	36.6	34.9	35.4	41.6	39.9	37.9	38.1	32.5	36.5	35.6	39.2	23.2	21.9
LLaVA-1.5-13B [70]	37.9	38.8	37.6	44.5	41.0	40.4	40.4	34.1	39.4	47.1	44.3	24.9	23.5
LLaVA-Onevision-7B [71]	23.4	23.8	23.4	31.3	27.8	18.4	21.3	21.2	21.3	31.3	21.7	13.6	12.0
Llama3.2-3B-vl [72]	25.3	25.6	23.4	27.8	25.4	30.3	28.9	27.8	25.0	30.3	35.8	7.3	7.0
Llama-3.2-11B-Vision-Instruct [72]	37.5	39.1	36.5	39.5	38.8	41.7	39.2	38.0	37.5	45.2	41.1	29.4	26.7
Llama-3.2V-11B-cot [72]	45.8	46.0	44.2	53.4	48.2	45.4	46.1	47.1	45.5	49.0	51.8	30.0	30.8
DeepSeek-VL2-Tiny [73]	29.1	34.2	31.4	38.8	35.4	36.7	32.1	33.7	35.4	38.5	32.2	19.8	19.2
Qwen2.5VL-7B [54]	38.4	41.9	28.7	44.8	42.8	40.8	39.7	49.4	43.2	44.2	41.2	24.9	24.8
Qwen3-VL-8B [74]	51.8	56.7	53.2	63.0	57.2	63.6	56.0	54.9	54.8	56.7	57.4	41.2	39.4
InternVL2.5-8B [61]	47.8	50.1	47.0	55.9	50.0	57.8	50.9	50.6	50.6	51.4	50.1	29.4	32.8
InternVL3-8B [62]	49.8	55.4	50.4	57.7	54.0	60.6	53.3	54.9	50.8	58.2	54.1	42.9	39.1
InternVL3.5-8B [75]	55.2	61.7	56.9	62.3	58.9	63.1	60.4	67.8	60.5	65.9	61.6	45.2	43.7
Open-source medical													
HealthGPT-M3 [21]	49.4	53.7	51.0	57.3	55.8	52.7	54.4	56.9	54.4	64.9	58.9	31.6	31.3
LLaVA-Med-v1.5-Mistral-7B [65]	17.9	20.9	18.6	24.6	21.3	22.5	20.5	21.6	21.5	25.0	21.4	7.3	7.7
HuatuoGPT-Vision-7B [66]	54.6	58.2	54.6	61.9	61.4	58.7	59.4	60.0	54.7	65.9	58.1	40.1	36.6
Patho-R1-3B [50]	58.4	63.8	57.9	67.6	61.1	66.5	64.1	67.5	62.5	74.5	72.2	36.7	35.0
Patho-R1-7B [50]	63.0	69.5	63.4	72.2	66.1	67.9	67.9	75.3	66.4	81.7	75.3	44.6	45.4
Quilt-LLaVA-v1.5-7B [32]	33.1	34.0	32.0	37.0	34.8	32.6	35.3	38.4	36.7	42.8	41.4	14.1	14.7
PathGen-LLaVA-13B [76]	58.2	61.9	58.8	62.3	59.2	59.6	58.5	62.4	58.5	68.3	63.8	55.9	56.2
PathLens-3B (Ours)	61.3	64.4	61.1	70.5	63.3	70.9	64.1	64.3	62.5	61.5	64.0	50.8	51.6

Table 3

Comparison of **PathLens** with other MLLMs on Quilt-VQA benchmark.

Model	Commonsense	Faith.-Step	Faith.-Token	Halluc.	Info.-Step	Miss Step	Reason Align	Redund	Rep.-Token	Sem. Cov.-Step	Overall
InternVL2-8B [77]	4.30	2.51	2.42	2.78	3.03	2.97	2.90	7.48	8.58	2.52	3.94
InternVL2.5-8B [61]	2.72	2.61	2.58	2.70	2.57	2.59	2.58	9.16	9.16	2.59	3.92
InternVL3-8B [62]	3.83	2.61	2.52	2.79	2.96	2.95	2.85	8.00	8.73	2.61	3.98
Llava-Onevision-7B [71]	2.10	1.63	1.59	1.69	1.73	1.72	1.69	8.46	8.88	1.62	3.10
LLaMA-3.2-11B-Vision-Ins. [72]	2.76	2.53	2.50	2.67	2.52	2.54	2.52	9.10	9.14	2.52	3.87
LLaMA-3.2V-11B-CoT [72]	3.54	1.98	1.89	2.62	2.30	2.39	2.22	7.96	8.97	1.99	3.56
HealthGPT-M3 [21]	2.99	2.06	1.85	2.58	2.03	2.10	1.89	9.11	9.36	1.10	3.59
Patho-R1-3B [50]	5.14	3.10	2.75	3.04	3.77	3.57	3.42	6.74	8.17	3.07	4.27
Patho-R1-7B [50]	4.60	2.72	2.50	2.70	3.33	3.23	3.06	6.65	8.32	2.72	3.97
PathLens-3B (Ours)	5.90	4.10	4.10	5.54	4.50	4.20	4.01	8.66	8.88	4.37	5.43

generalist models such as InternVL3.5-8B (56.01%) and Qwen3-VL-8B (56.56%), highlighting that scaling model size alone does not guarantee robustness to pediatric-specific domain shifts. Furthermore, compared with other medical-specific MLLMs, including Huatuo-Vision-7B (58.96%) and Lingshu-7B (58.04%), PathLens demonstrates superior transferability. We attribute this advantage to our model's effective alignment strategy, which likely prevents overfitting to adult-dominant training data and enables the capture of intrinsic, generalized histological features. While a performance gap remains relative to proprietary systems such as Gemini 3 Pro, PathLens offers a highly efficient and accessible alternative, successfully bridging the generalization gap in resource-constrained pediatric pathology scenarios.

4.5. Ablation study

To understand the contribution of each training component, we conduct an ablation study comparing three configurations: *SFT-only*, *RLVR-only*, and the full *PathLens* pipeline that integrates both SFT and RLVR. As summarized in Table 5, the full PathLens model achieves the best performance across all benchmarks, improving PathVQA accuracy from 50.1/50.2% to 53.2% and boosting cross-domain performance on

PedPathVQA from 55.1/52.1% to 60.1%. These results indicate that SFT provides essential pathology-specific CoT priors, while RLVR further refines answer faithfulness and step-level completeness, yielding stronger and more robust clinical reasoning.

Fig. 7 analyzes the distribution of output token lengths under different training strategies. SFT-only produces the shortest responses, RLVR-only moderately increases reasoning length, and the full PathLens model generates the longest chains, reflecting richer and more explicit reasoning processes. Together, these findings suggest that combining high-quality pathology CoT supervision with targeted RL refinement is critical for eliciting detailed, clinically aligned reasoning, while simultaneously improving generalization across diverse pathology VQA benchmarks.

4.6. Expert validation

To rigorously assess the clinical reliability and reasoning fidelity of our framework, we conducted a two-phase expert evaluation: (1) assessment of the Patho-CoT data pipeline, and (2) comparison of PathLens with a strong baseline in terms of CoT reasoning quality.

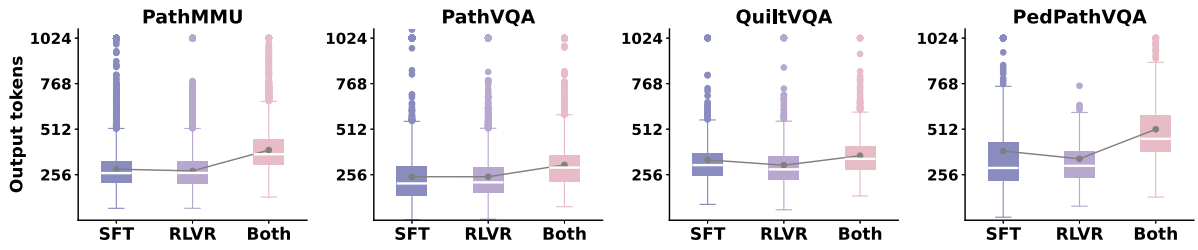


Fig. 7. Distribution of output token lengths across four benchmarks under different training strategies (SFT, RLVR, Both). Boxplots show medians, interquartile ranges, and outlier distributions.

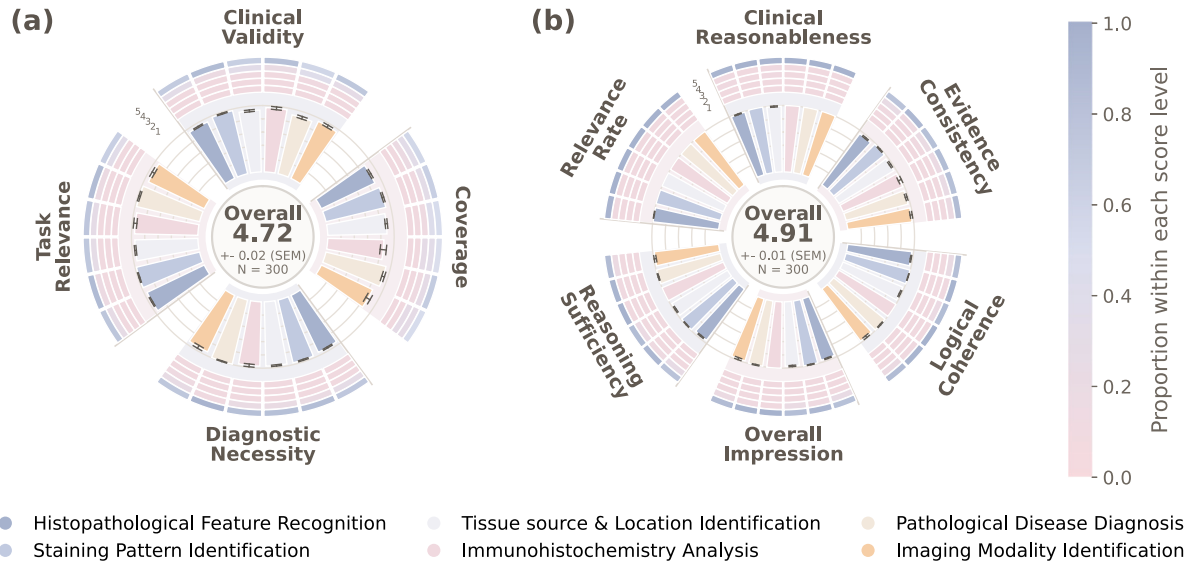


Fig. 8. Expert validation of Patho-CoT. (a) Multi-dimensional 5-point Likert-scale assessment of the Visual Feature Agent outputs across four criteria. (b) Multi-dimensional 5-point Likert-scale assessment of the final generated CoT across six criteria. Radial stacked bars show the proportion of scores from 1 to 5 within each question type, and the center reports the overall mean $\pm$ SEM over 300 evaluated cases.

Table 4
Comparison of **PathLens** with other MLLMs on Ped-PathVQA benchmark.

Type	Model	Acc
Proprietary	Claude Sonnet 4.5	50.09
	GPT-5.1	78.74
	Gemini 3 Pro	67.47
Open-source general	InternVL3-8B	58.23
	InternVL3.5-8B	56.01
	Qwen2.5-VL-7B	50.83
	Qwen3-VL-8B	56.56
Open-source medical	Med-R1-2B	38.82
	MedVLM-R1-2B	43.81
	HealthGPT-M3	45.10
	Lingshu-7B	58.04
	Huatuo-Vision-7B	58.96
	Patho-R1-3B	54.16
	Patho-R1-7B	58.78
	PathLens-3B (Ours)	60.07

The evaluation was conducted by two junior pathologists, with each case reviewed by one and subsequently audited by a senior pathologist to ensure consistency, resolve discrepancies, and maintain clinical validity. All assessments were conducted using a standardized 5-point Likert scale [78]. Detailed criteria and the evaluation interface are provided in [Appendix A](#).

In the first phase, experts evaluated both the intermediate Visual Feature Agent outputs and the final synthesized CoT, with results shown in [Fig. 8](#). Patho-CoT achieves consistently high scores across all dimensions, with particularly strong performance in clinical validity and clinical reasonableness. This suggests that grounding reasoning in visual features effectively mitigates hallucinations and aligns with expert diagnostic workflows, resulting in high-fidelity multimodal supervision.

In the second phase, experts blindly compared reasoning outputs from PathLens and Patho-R1-7B using the same evaluation protocol. As shown in [Fig. 9](#), PathLens achieves competitive overall performance despite having less than half the parameters, and notably outperforms in logical coherence and reasoning sufficiency. This demonstrates that the ReAL strategy effectively enforces structured, evidence-grounded reasoning without relying on model scaling.

Overall, the expert study provides human-centered validation for both the data and model components, confirming that Patho-CoT yields clinically reliable supervision, enabling PathLens to develop more interpretable and clinically grounded reasoning capabilities.

4.7. Case study

To illustrate model behavior, we analyze a representative case involving PAX8 immunohistochemical staining, as shown in [Fig. 10](#). This task is challenging, as multiple candidate answers may be plausible, requiring the model to correctly interpret the staining pattern and select an appropriate level of diagnostic specificity.

PathLens correctly identifies positive PAX8 staining and associates it with tumors of renal origin, leading to the conclusion that a positive

Table 5
Ablation study of different training strategies on multiple pathology VQA benchmarks.

Method	PathVQA	PathMMU-Val	PathMMU-Test	PedPathVQA	Mean Token Nums
SFT-Only	50.1	56.2	55.2	55.1	278.4
RLVR-Only	50.2	55.0	55.1	52.1	269.9
Both (Full PathLens)	53.2	61.3	61.1	60.1	366.0

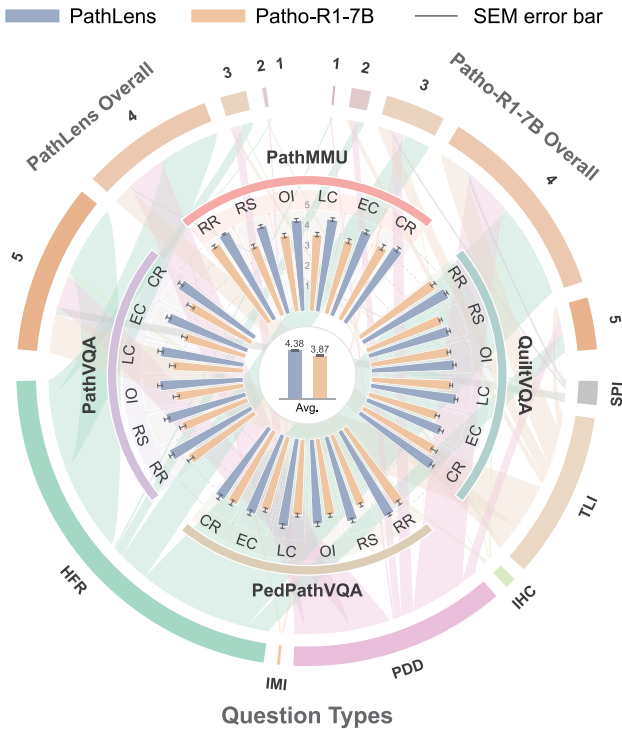


Fig. 9. Expert comparison of benchmark reasoning quality between PathLens and Patho-R1-7B. Pathology experts rated model-generated reasoning on six CoT criteria using the same 5-point Likert scale: RR, relevance rate; RS, reasoning sufficiency; LC, logical coherence; EC, evidence consistency; CR, clinical reasonableness; and OI, overall impression. Blue and orange bars denote the mean scores of PathLens and Patho-R1-7B, respectively, with SEM error bars. The outer rings summarize benchmark and question-type groupings, and the center shows the overall average expert score of each model. Question types are abbreviated as follows: HFR, Histopathological feature recognition; PDD, Pathological disease diagnosis; TLI, Tissue source & location identification; SPI, Staining pattern identification; IHC, Immunohistochemistry analysis; and IMI, Imaging modality identification.

result supports the diagnosis of kidney cancer. Its reasoning remains grounded in visual evidence, avoiding overly specific interpretations. In contrast, baseline models often produce over-specific conclusions, unstable reasoning, or insufficient justification. This case highlights that effective pathology VQA requires not only correct predictions but also evidence-consistent reasoning at an appropriate level of diagnostic granularity. Additional qualitative examples are provided in [Appendix B](#), further illustrating the reasoning behavior of PathLens across diverse pathology scenarios.

5. Discussion

In this study, we introduce PathLens, a compact and highly efficient VQA model designed for pathology, which aims at addressing the reasoning disparity between lightweight architectures and large-scale multimodal systems within the professional clinical domain. The experimental results validate our hypothesis that high-quality, domain-specific supervision, when combined with a structured alignment process, can yield robust diagnostic reasoning capabilities. In this section,

we will thoroughly examine the effectiveness of the model, analyze the inherent mechanisms of the proposed approach, and objectively evaluate the limitations of the current work.

5.1. Effectiveness and out-of-domain generalization

PathLens, with a parameter scale of 3B, has demonstrated outstanding performance in multiple benchmark tests, such as PathVQA, PathMMU, Quilt-VQA, and PedPathVQA. Its robustness is fundamentally attributed to the effective fine-tuning of the medical foundation model with specialized pathological data. By utilizing the ReAL framework, the model can efficiently acquire domain-specific reasoning patterns without the necessity of pre-training from scratch. These findings verify that high-quality, pathology-specific supervision can successfully bridge the performance gap between lightweight architectures and larger multimodal systems on reasoning-intensive clinical tasks.

It should be noted that, even though the training data is sourced entirely from adult cases, PathLens still maintains its competitiveness in pediatric benchmark tests. This indicates that the learned supervisory signals capture transferable morphology-to-diagnosis mappings rather than relying on surface features specific to a particular distribution, thereby demonstrating excellent cross-domain generalization capabilities.

5.2. Role of the Patho-CoT pipeline

The Patho-CoT pipeline serves a deliberate function within the overall framework by supplying pathology-specific descriptive patterns, key diagnostic terminology, and a structured organization of evidence and conclusions. Through this pipeline, the synthesized training data enforces a “reason-before-answering” pattern and maintains explicit links between visual observations and diagnostic statements. Specifically, the diverse dataset composition and extensive token lengths illustrated in [Fig. 11](#) demonstrate that the synthesized sequences provide substantial and detailed supervisory signals. Concurrently, the vocabulary distribution in [Fig. 12](#) validates the strong concentration of clinical terminology within the corpus. These semantic context ensure that the training data establishes a robust, domain-specific reasoning prior, which directly enhances the capacity of the model to generate organized and clinically interpretable responses.

To guarantee the clinical reliability of the synthetic data, we have implemented a multi-layer quality control mechanism. Every component within the multi-agent collaboration framework has undergone domain-specific prompt engineering and constraints to ensure that the visual feature extraction and reasoning generation adhere to pathological norms. Manual sampling evaluations indicate that more than 98% of the synthetic CoT data meet professional standards in terms of clinical logic and terminology usage (scoring above 4.5 out of 5).

5.3. Mechanisms of the ReAL framework

Building upon the structured prior established by Patho-CoT, the distinction between the two training stages of the ReAL framework is essential to elucidate the behavior of the final model. Specifically, the GSPO mechanism functions at the response level, rewarding the accuracy and adherence to the format of the final answer instead of offering supervision for each intermediate step of the reasoning process. Although rewards based solely on the final outcome may occasionally cause the model to generate spurious reasoning to attain the correct

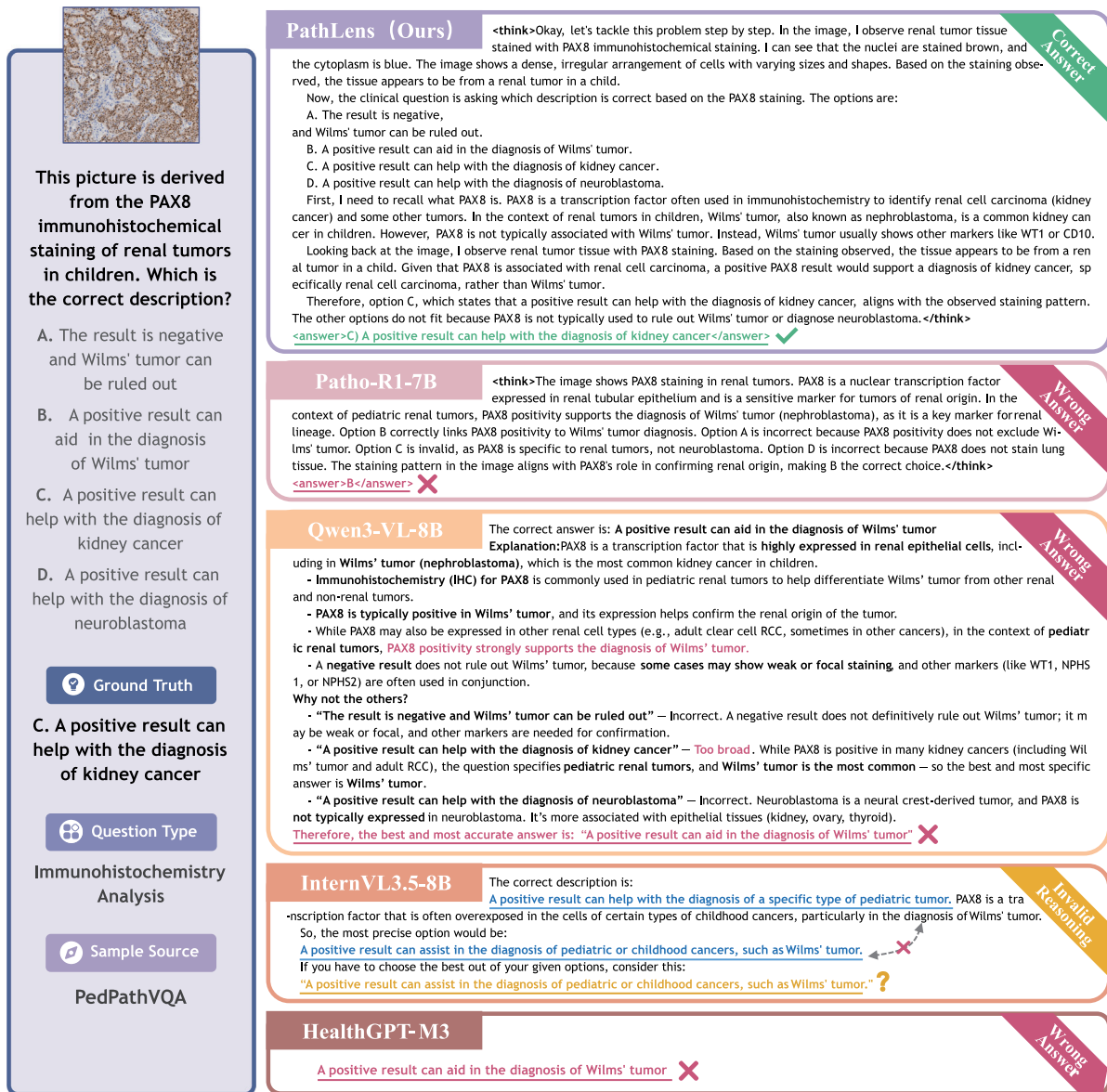


Fig. 10. Qualitative case study on a pediatric renal tumor immunohistochemistry sample from PedPathVQA. Given a PAX8-stained image and a multiple-choice question, PathLens selects the ground-truth answer and generates an evidence-grounded reasoning chain that links the observed positive staining pattern to renal tumor diagnosis at an appropriate level of specificity. In contrast, Patho-R1-7B and Qwen3-VL-8B over-specify the conclusion as Wilms' tumor, InternVL3.5-8B produces invalid and semantically unstable reasoning, and HealthGPT-M3 provides only a terse answer without an explicit rationale. This example highlights that PathLens better balances visual evidence, pathology knowledge, and answer granularity in challenging pediatric pathology VQA.

conclusion, the ReAL framework alleviates this problem by implementing reinforcement learning only after a rigorous supervised fine-tuning phase.

Through comprehensive supervised fine-tuning on the high-quality CoTs generated by Patho-CoT, the model has internalized the correct diagnostic reasoning patterns. In the first stage of supervised learning, the model's reasoning space is strictly restricted to clinically reasonable paths. As a result, when reinforcement learning is applied in the second stage, the action space is significantly reduced, substantially reducing the likelihood of generating unreasonable inferences. Ablation experiments further validate this mechanism: models utilizing only GSPO without prior supervised fine-tuning do exhibit a deterioration in reasoning quality, while the complete two-phase training maintains clinical consistency in reasoning, owing to the essential reasoning framework provided by the alignment with supervised CoTs, and reinforcement learning serves as a crucial optimization mechanism

that suppresses undesirable outputs without diminishing the clinical effectiveness of intermediate steps.

5.4. Model positioning and clinical practicality

Compared to proprietary systems such as GPT-4.1 and Gemini-2.5, the performance gap of PathLens-3B is approximately 2 percentage points. Nevertheless, this minor difference is of secondary importance under the practical constraints of clinical deployment. Proprietary models rely on cloud inference, which necessitates the transmission of patient histological images to external servers. This poses significant privacy risks and regulatory challenges within the frameworks of HIPAA [55] and GDPR [56]. In contrast, the core advantage of PathLens lies in the equilibrium between parameter efficiency and inference quality. It substantially reduces the demand for computing resources while maintaining a reasonable accuracy rate, and can be fully deployed locally on consumer-grade hardware, as detailed in Appendix E.

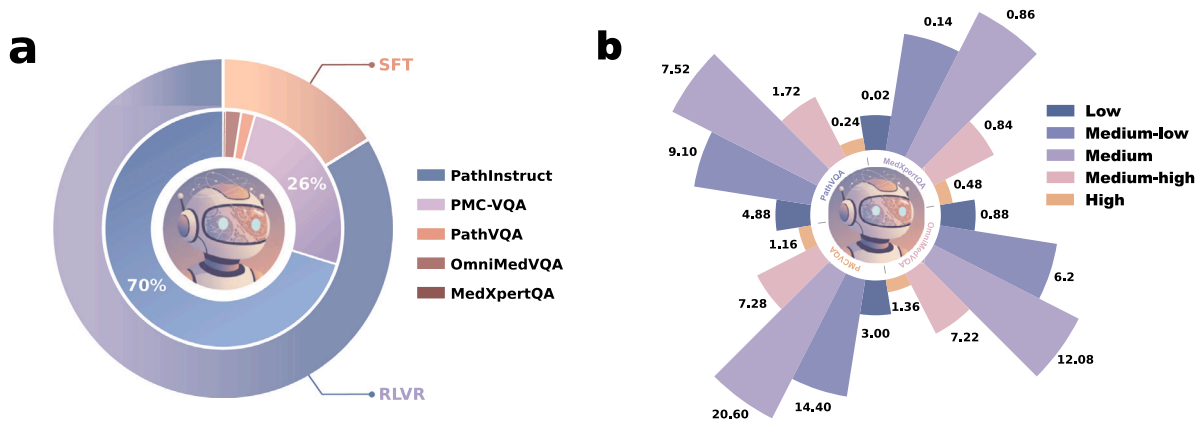


Fig. 11. Dataset visualization. (a) Subset proportions in the dataset (inner ring) and their use in ReAL (outer ring). (b) Token length distribution of CoT data across four subsets.

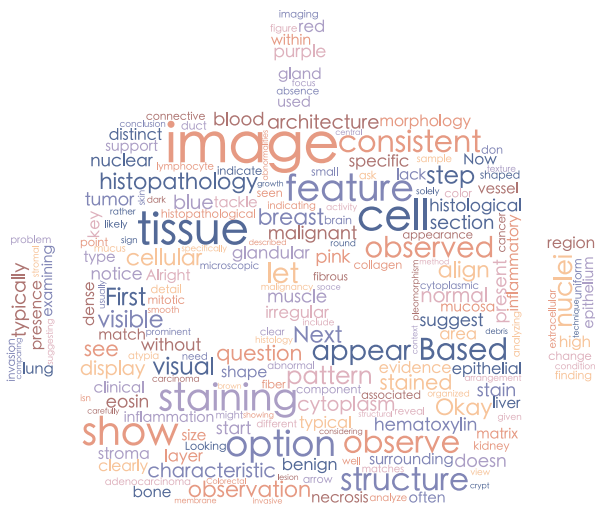


Fig. 12. Word cloud of CoT tokens from the dataset used in ReAL Stage I. Frequently occurring terms highlight the emphasis on image-based reasoning, tissue morphology, and cellular features within the synthetic reasoning corpus.

Therefore, PathLens achieves an optimized balance between diagnostic accuracy, resource constraints, and data sovereignty.

In the open-source domain, Patho-R1-7B achieves a higher overall accuracy on PathMMU (63.0% compared to PathLens’s 61.3%) and demonstrates a significant advantage on its Atlas subset (75.3% compared to 61.5%). This difference primarily stems from the disparity in parameter scale and training data volume: Patho-R1 employs millions of image-text pairs, including a large quantity of pathology textbook data, which gives it an advantage in retrieval tasks. PathLens, contrastly, is designed to maximize clinical inference capabilities under strict resource constraints, achieving efficient training through a highly curated, compact dataset. Despite the performance gap on the encyclopedic subset, PathLens remains a highly practical architecture, offering strong diagnostic inference capabilities for resource-constrained scenarios.

5.5. Limitations

While the current architecture demonstrates robust performance, several constraints warrant acknowledgment. First, the quality of the

reasoning chains synthesized by Patho-CoT remains bounded by the inherent capabilities of the underlying LLMs despite the implementation of multi-layered quality control. This inevitably restricts the semantic depth and diversity of the generated training corpus. Second, the current reward mechanism focuses primarily on final answer correctness and structural compliance. There remains significant potential to incorporate fine-grained supervision of intermediate reasoning steps to further enhance the fidelity and clinical grounding of the model’s logical process. Finally, although VQA benchmarks provide a rigorous assessment of reasoning capacity, they do not fully capture the multifaceted complexity of real-world clinical workflows. These limitations reflect broader challenges within the field of AI-driven pathology and delineate clear trajectories for future research.

6. Conclusion and future work

This paper presented PathLens, a lightweight yet capable multi-modal pathological reasoner designed to deliver clinically faithful and interpretable VQA. Its key contribution lies in addressing the challenge of integrating fine-grained visual cues with structured logic rather than producing direct predictions. To the best of our knowledge, this work is the first to leverage a dedicated multi-agent pipeline, Patho-CoT, to synthesize high-quality multimodal CoT reasoning by unifying visual cue hypothesis and clinically grounded description enhancement. Our study shows that the proposed ReAL strategy, which couples this synthetic CoT supervision with GSPO-based RL, markedly improves reasoning fidelity, structural consistency, and clinical plausibility, enabling efficient generation of transparent diagnostic sequences. To support rigorous evaluation, we introduced PedPathVQA, the first pilot dataset dedicated to pediatric pathology VQA, and empirical results demonstrate that PathLens achieves substantial gains over competitive medical and general-purpose MLLMs while maintaining exceptional parameter efficiency.

This paper presented PathLens, a lightweight yet capable multi-modal pathological reasoner designed to deliver clinically faithful and interpretable VQA. Its key contribution lies in integrating fine-grained visual cues with structured diagnostic reasoning, moving beyond one-off prediction. To our knowledge, this is the first work to leverage a dedicated multiagent pipeline, Patho-CoT, to synthesize high-quality multimodal CoT reasoning by unifying visual cue hypotheses with clinically grounded description enhancement. Our study shows that the proposed ReAL strategy, which couples synthetic CoT supervision with GSPO-based RL, significantly improves reasoning fidelity, structural

consistency, and clinical plausibility, enabling the efficient generation of transparent diagnostic sequences. To support rigorous evaluation, we introduced PedPathVQA, the first pilot dataset dedicated to pediatric pathology VQA. Empirical results show that PathLens achieves substantial gains over competitive medical and general-purpose MLLMs while maintaining exceptional parameter efficiency.

Despite the demonstrated success of PathLens, the transition from benchmark performance to bedside utility requires further exploration. Our future work will prioritize the integration of larger-scale, high-fidelity datasets, with a specific focus on expanding PedPathVQA to support the pediatric pathology community. Moving forward, transitioning the model toward high-complexity tasks such as whole-slide image analysis and comprehensive report synthesis will better illustrate its value in real-world clinical settings. Perhaps most crucially, advancing reinforcement learning to evaluate the ‘process’ of reasoning, rather than just the final output, is a high-priority objective. Crafting reward signals that reflect the subtle expertise inherent in clinical diagnosis will be vital for ensuring that model logic truly mirrors that of a human pathologist.

CRediT authorship contribution statement

Zhu Zhu: Writing – original draft, Project administration, Methodology, Conceptualization. **Huangwei Chen:** Writing – original draft, Software, Methodology, Formal analysis. **Zhenyu Yan:** Software, Methodology, Investigation, Conceptualization. **Donghao Zhang:** Visualization, Software, Investigation. **Yueyi Wu:** Visualization, Validation, Investigation. **Yuqi Zhan:** Visualization, Validation, Data curation. **Weihao Cheng:** Investigation. **Manli Zhao:** Data curation. **Weizhong Gu:** Data curation. **Yifei Chen:** Writing – review & editing. **Feiwei Qin:** Writing – review & editing, Supervision, Resources, Funding acquisition. **Gang Yu:** Writing – review & editing, Supervision, Resources, Funding acquisition.

Ethical approval

The study was conducted in accordance with the Declaration of Helsinki (as revised in 2013) and approved by the Academic Ethics Committee of Children’s Hospital Zhejiang University School of Medicine (No. 2023-IRB-0287-P-01). Individual consent for this retrospective analysis was waived.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table A.6
General 5-point Likert scale used in expert evaluation.

Score	Rating	Definition
1	Very poor	Completely unsatisfactory; content is irrelevant, incorrect, severely incomplete, or inconsistent with image and clinical evidence.
2	Poor	Substantially flawed; major problems in relevance, logic, completeness, or clinical correctness, with limited useful information.
3	Fair	Partially acceptable; generally related to the task, but with noticeable weaknesses, omissions, or inconsistencies that reduce reliability.
4	Good	Mostly satisfactory; relevant, logically sound, and largely consistent with image evidence and clinical reasoning, with only minor issues.
5	Excellent	Fully satisfactory; highly relevant, complete, logically coherent, evidence-grounded, and clinically consistent.

Appendix A. Expert validation details

To ensure the clinical reliability and interpretability of the synthesized outputs, we incorporate an expert validation protocol into the Patho-CoT pipeline. Board-certified pathologists evaluate a subset of generated results, including visual feature extraction and final CoT reasoning, using a standardized 5-point Likert scale (Table A.6) to ensure consistent and comparable scoring.

For the Visual Feature Agent, evaluation focuses on task relevance, diagnostic necessity, clinical validity, and coverage (Table A.7), ensuring that extracted features are both medically meaningful and diagnostically informative. For CoT reasoning, we adopt a comprehensive rubric (Table A.8) covering overall quality, relevance, logical coherence, reasoning sufficiency, consistency with image evidence, and clinical plausibility, thereby assessing both reasoning quality and medical correctness.

As shown in Fig. A.13, the expert validation protocol was deployed through a dedicated scoring and review workbench with role-specific access for junior and senior pathologists. The system provides a unified workflow spanning the portal, case overview dashboard, and specialized grading dashboards for Visual Feature Agent outputs, synthesized CoT samples, and PathLens output CoT. This design standardizes case assignment, score recording, and hierarchical review, thereby improving both evaluation efficiency and consistency across annotators.

Fig. A.14 further illustrates the detailed case-level grading interfaces used in the validation process. For each sample, the interface presents the pathology image together with the generated textual output and a structured scoring panel aligned with the corresponding evaluation rubric. This layout enables experts to assess visual feature extraction and reasoning quality in a fine-grained and evidence-aware manner, while also facilitating direct comparison between junior-pathologist assessments and senior-pathologist auditing results.

Appendix B. Supplementary case studies

In addition to the representative example shown in the main text, we further provide three supplementary qualitative case studies in Figs. B.15–B.17 to illustrate the reasoning characteristics of PathLens under different pathology VQA settings, including histopathological feature recognition, disease diagnosis, and tissue source identification. Across these cases, we compare PathLens with several representative baselines, namely Patho-R1-7B, Qwen3-VL-8B, InternVL3.5-8B, and HealthGPT-M3. The comparisons consistently show that the advantage of PathLens does not lie merely in producing the correct final option, but in generating reasoning traces that are more visually grounded, diagnostically targeted, and better aligned with the granularity required by the question.



Fig. A.13. Overview of the expert validation system. The figure presents the role-specific interfaces for Junior Pathologist and Senior Pathologist, including the login portal, overview dashboard, Visual Feature Agent grading dashboard, synthesized CoT grading dashboard, and PathLens output CoT grading dashboard, illustrating the standardized workflow and auditing model-generated pathology reasoning outputs.

Case 1: Calcification pattern recognition (Fig. B.15). The first example is drawn from PathMMU and focuses on fine-grained recognition of calcification morphology. Although several distractor options appear superficially plausible, the correct answer is *coarse, granular focal calcification*. PathLens arrives at the correct prediction by explicitly grounding its reasoning in the focal distribution and irregular granular appearance of the deposit, while also ruling out unsupported alternatives such as vascular linear calcification and lamellar concentric-ring structures. This behavior is diagnostically meaningful because the task depends on recognizing subtle morphological distinctions rather than recalling a generic calcification concept. In contrast, the compared baselines exhibit characteristic failure modes. Patho-R1-7B over-associates the

lesion with vascular-oriented calcification, whereas Qwen3-VL-8B and InternVL3.5-8B are drawn toward a more prototypical but visually unsupported lamellar pattern. HealthGPT-M3 outputs only a short, incorrect answer without an interpretable rationale. This example therefore shows that PathLens is better at preserving local visual fidelity when multiple answer candidates share overlapping clinical semantics.

Case 2: Papillary thyroid carcinoma assessment (Fig. B.16). The second case, from QuiltVQA, asks whether the nuclear features qualify as papillary thyroid carcinoma. Here, the correct response is yes, but the challenge lies in determining whether the observed morphology is sufficient for a clinically meaningful affirmative judgment. PathLens



Fig. A.14. Examples of the detailed grading interfaces used in expert validation. For both Junior Pathologist and Senior Pathologist, the interface displays the pathology image, generated outputs, and structured scoring panels for evaluating Visual Feature Agent outputs, synthesized CoT, and PathLens output CoT, supporting fine-grained and rubric-based assessment of reasoning quality and clinical reliability.

first identifies the key visual evidence, including papillary structures with fibrovascular cores, nuclear enlargement, grooves, and inclusions, and then translates these observations into a direct diagnostic conclusion. Importantly, its reasoning remains concise and decisive, which is desirable in pathology VQA settings where the final answer space is narrow and clinically specific. By contrast, Patho-R1-7B produces a partially correct yet over-qualified explanation that shifts attention toward additional subtype criteria, thereby weakening the directness of

the answer. Qwen3-VL-8B contains noticeable semantic inconsistency, initially arguing against the diagnosis before later acknowledging evidence compatible with papillary thyroid carcinoma. InternVL3.5-8B makes an outright negative error, while HealthGPT-M3 gives a relevant answer but without sufficient interpretability. This case highlights that PathLens is not only more accurate but also better calibrated in how much evidence is needed to support a binary pathological judgment.

Table A.7

Evaluation rubric for Visual Feature Agent outputs.

No.	Dimension	Evaluation focus
1	Task Relevance	Whether extracted visual features directly support answering the target question.
2	Diagnostic Necessity	Whether extracted features are diagnostically essential or highly informative for differentiating the target disease/entity.
3	Clinical Validity	Whether extracted features are medically meaningful and consistent with established pathological knowledge.
4	Coverage	Whether the VF Agent captures major discriminative features and key differential clues without omitting critical evidence.

Case 3: Gastrointestinal tissue identification (Fig. B.17). The third example comes from PathVQA and evaluates whether gastrointestinal tissue is present in the image. Unlike the previous two cases, this task emphasizes broad tissue recognition rather than lesion-specific diagnosis. PathLens correctly answers *yes* by integrating multiple histological landmarks, including gastric pits and glands, intestinal villi and crypts, and stratified squamous epithelium. The key strength here is compositional reasoning: instead of relying on a single cue, the model accumulates several tissue-level observations and uses them jointly to support the final conclusion. In contrast, Patho-R1-7B hallucinates an unrelated squamous-cell-carcinoma immunohistochemistry context and therefore rejects the correct answer. Qwen3-VL-8B and InternVL3.5-8B also fail, largely because they default to uncertainty or generic exclusion reasoning when explicit contextual labels are unavailable. HealthGPT-M3 again produces only a terse answer with limited explanatory value. This example suggests that PathLens is better at transforming heterogeneous visual cues into a coherent tissue-level interpretation, which is especially important for open-ended pathology VQA scenarios.

Taken together, these supplementary case studies reaffirm that by strategically fine-tuning the HealthGPT foundation model, PathLens successfully elicits explicit and reliable CoT reasoning without compromising final-answer accuracy. Crucially, this demonstrates that achieving high-fidelity clinical interpretability does not strictly require computationally exhaustive pre-training from scratch. Instead, our highly efficient adaptation strategy leverages targeted, domain-specific supervision to unlock robust diagnostic logic within a lightweight architecture, effectively bridging the gap between parameter efficiency and trustworthy clinical decision-making.

Appendix C. Supplementary details of the PedPathVQA dataset

To provide a comprehensive overview, we present the detailed statistical distribution of our PedPathVQA dataset in Table C.9. The dataset encompasses pediatric diseases of clinical significance that require pathological examination, including pediatric tumors, inflammatory and infectious diseases, immune diseases, congenital and developmental abnormalities, and benign lesions with reactive hyperplasia.

- **Neoplastic Lesions (41.4%):** The dataset comprises 24 types of neoplastic lesions, representing the largest category. This aligns with the core diagnostic challenge in pediatric pathology — the differential diagnosis of tumors. It includes common childhood malignancies such as neuroblastoma (6.5%) and Wilms tumor (6.5%), ensuring representativeness in critical diagnostic tasks.

- **Inflammation & Infection (16.0%):** This category includes 9 types of inflammatory diseases (e.g., pneumonia, gastritis). This is crucial for evaluating a model’s ability to distinguish cancer from inflammation, preventing reliance on tissue necrosis or cellular atypia for coarse judgments.
- **Immune Diseases (16.0%):** This category includes 4 types of immune-related conditions. Including these disorders ensures the model can recognize specific immune-mediated tissue damage patterns distinct from infections or neoplasms.
- **Congenital & Developmental Abnormalities (5.6%):** This category includes developmental abnormalities such as Hirschsprung’s Disease and Biliary Atresia, capturing the unique spectrum of pediatric disease.
- **Benign Lesions & Reactive Hyperplasia (21.0%):** the dataset contains 14 types of benign lesions and reactive hyperplasia. These cases represent high-risk areas for clinical misdiagnosis; thus, the model must demonstrate excellent fine-grained recognition to differentiate them from malignant tumors or complex inflammations.

In summary, PedPathVQA is not merely a classification dataset but a VQA benchmark that requires the model to understand pathological physiological processes, such as tumor growth, immune attack, or inflammatory repair. The model must provide accurate answers based on visual features in diverse contexts to achieve optimal performance.

Appendix D. RL hyperparameter

The hyperparameter configuration used for the second-stage ReAL (GSPO) training is summarized in Table D.10.

Appendix E. Inference efficiency analysis

To further substantiate the lightweight nature of PathLens, we conduct an inference efficiency analysis on the PedPathVQA benchmark. All models are evaluated under the same hardware and inference backend settings, using a single NVIDIA RTX 3090 GPU and the same inference stack based on Hugging Face Transformers with PyTorch, FP16 precision, and greedy decoding. We report two deployment-oriented metrics: end-to-end generated-token throughput and peak GPU memory usage. The throughput is measured by tokens per second (TPS), computed as the number of generated tokens divided by the total wall-clock inference time, including image encoding, prompt prefix, and autoregressive decoding.

As shown in Table E.11, PathLens achieves the highest throughput among the compared models, reaching 23.02 TPS, while requiring only 8.89 GB peak GPU memory. Compared with larger 7B/8B baselines, PathLens maintains comparable or slightly higher throughput with substantially lower memory consumption. Specifically, PathLens reduces peak GPU memory usage by 44.2% compared with Patho-R1-7B, and by 48.0% and 48.2% compared with Qwen3-VL-8B and InternVL3.5-8B, respectively. These results indicate that PathLens is lightweight not only in terms of parameter scale, but also in terms of GPU memory footprint, supporting its suitability for low-cost local deployment in privacy-sensitive clinical environments.

Data availability

Data will be made available on request.

Table A.8
Evaluation rubric for CoT.

No.	Dimension	Evaluation focus
1	Overall Impression	Overall quality of the generated CoT, including whether it appears trustworthy, useful, and well-formed as a diagnostic reasoning process.
2	Relevance Rate	Degree to which the reasoning directly addresses the target question, without off-topic or redundant content.
3	Logical Coherence	Internal consistency and organization of reasoning steps, and whether each step meaningfully advances the diagnosis.
4	Reasoning Sufficiency	Sufficiency of intermediate reasoning and diagnostic evidence to support the final conclusion.
5	Consistency with Image Evidence	Consistency of the reasoning with observed pathology image evidence, without unsupported or hallucinated findings.
6	Medical Accuracy/Clinical Plausibility	Alignment with pathological knowledge, clinical standards, and realistic diagnostic logic.

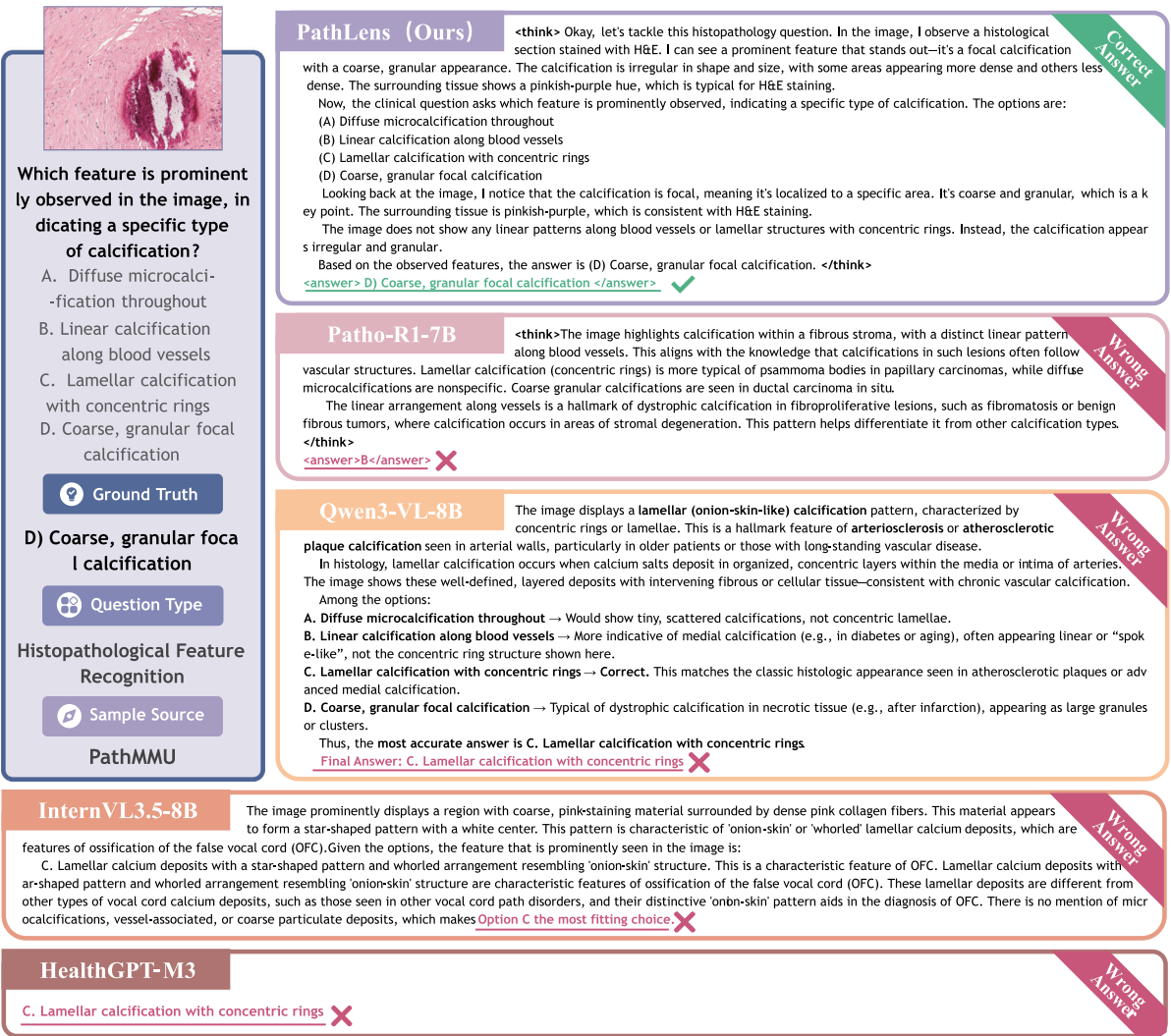


Fig. B.15. The case asks the model to identify the dominant calcification pattern from several visually similar choices. PathLens correctly predicts *coarse, granular focal calcification* by grounding its reasoning in the localized, irregular, and non-lamellar morphology of the deposit. In contrast, Patho-R1-7B incorrectly interprets the lesion as vascular-associated linear calcification, while Qwen3-VL-8B, InternVL3.5-8B, and HealthGPT-M3 drift toward a prototypical but unsupported lamellar/concentric-ring interpretation. This example highlights that PathLens is better able to distinguish subtle morphological differences and avoid over-reliance on stereotypical calcification templates.

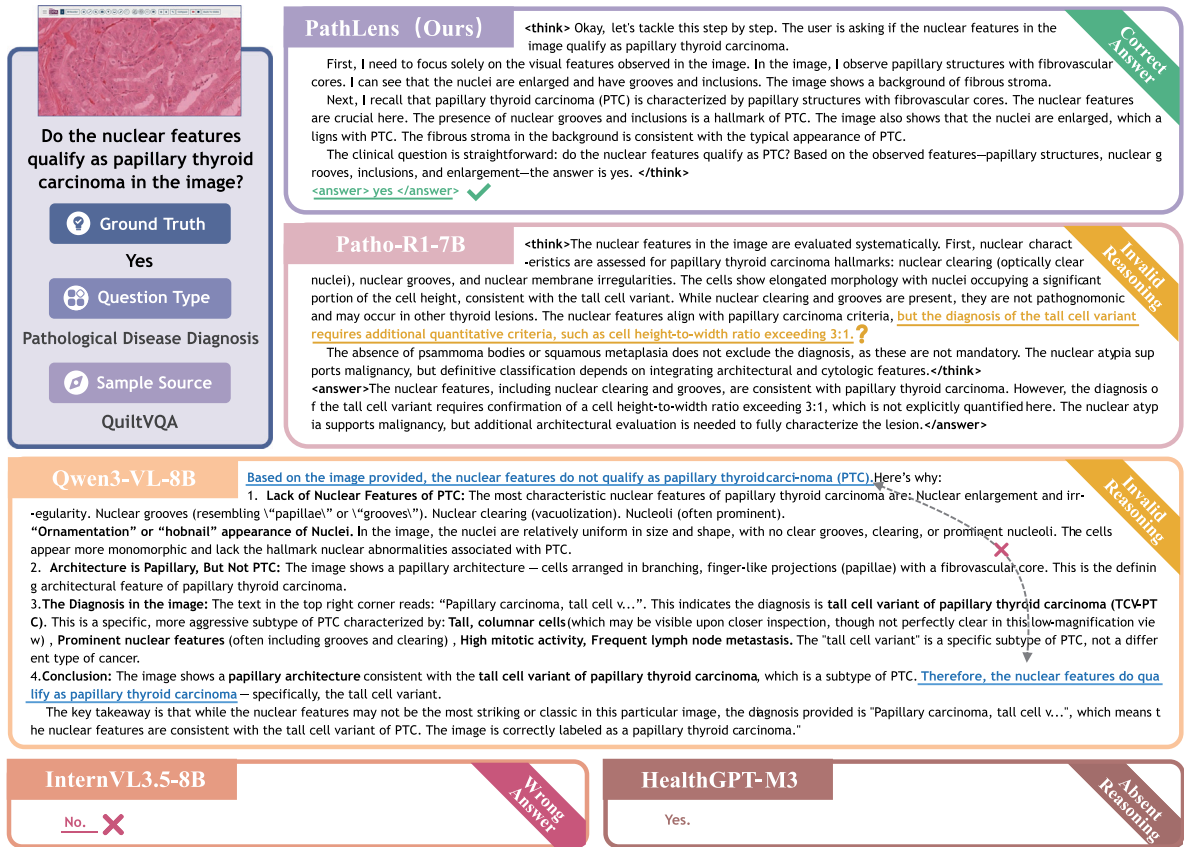


Fig. B.16. This example evaluates whether the nuclear features in a thyroid pathology image are sufficient to support papillary thyroid carcinoma. PathLens gives the correct affirmative answer and supports it with a concise, evidence-grounded reasoning chain based on papillary architecture, nuclear enlargement, grooves, and inclusions. By comparison, Patho-R1-7B produces a partially correct but overly qualified explanation, Qwen3-VL-8B shows internal inconsistency before arriving at a conclusion, InternVL3.5-8B makes an erroneous negative judgment, and HealthGPT-M3 provides only a short relevant answer without an interpretable rationale. The case illustrates that PathLens achieves a better balance between correctness, decisiveness, and transparency in clinically meaningful diagnostic reasoning.

Table C.9
Classification of diseases contained in the PedPathVQA dataset based on their nature.

Disease category	Diseases
Neoplastic Lesions (24 diseases)	Neuroblastoma (35), Wilms Tumor (35), Lymphoma (21), Tonsillar Lymphoma (20), Rhabdomyosarcoma (18), Glioma (17), Clear Cell Sarcoma of the Kidney (10), Ependymoma (9), Renal Cell Carcinoma (8), Hepatoblastoma (7), Meningioma (6), Schwannoma (6), Adrenocortical Neoplasms (5), Epithelioid Hemangioendothelioma (4), Medulloblastoma (3), Ewing Sarcoma of the Kidney (3), Lymphatic Metastasis of Neoplasm (3), Pilocytic Astrocytoma (3), Teratoma (3), Gastrointestinal Tumors (2), Osteochondroma (2), Craniopharyngioma (2), Yolk Sac Tumor (1), Melanoma (1)
Inflammation & Infection (9 diseases)	Gastritis (23), Enteritis (22), Cholestatic Liver Disease (9), Nephritis (9), Necrotizing Lymphadenitis (8), Other Chronic Inflammation (7), Tonsillitis (4), Active Esophagitis (3), Pneumonia (2)
Immune Diseases (4 diseases)	IgA Nephropathy (30), Systemic Lupus Erythematosus (14), Purpura (19), Glomerular Diseases (30)
Congenital & Developmental Abnormalities (3 diseases)	Hirschsprung's Disease (18), Cerebral Aneurysm (9), Biliary Atresia (2)
Benign Lesions & Reactive Hyperplasia (14 diseases)	Normal Tissue (29), Hemangioma (16), Xanthogranuloma of Infancy (12), Nodular Fasciitis of the Back (10), Compound Nevus (9), Junctional Nevus (6), Pilomatricoma (6), Epidermal Verrucous Nevus (5), Langerhans Cell Histiocytosis (5), Neurofibroma (5), Intradermal Nevus (4), Myositis (4), Reactive Hyperplasia of Lymph Nodes (2), Cyst (1)

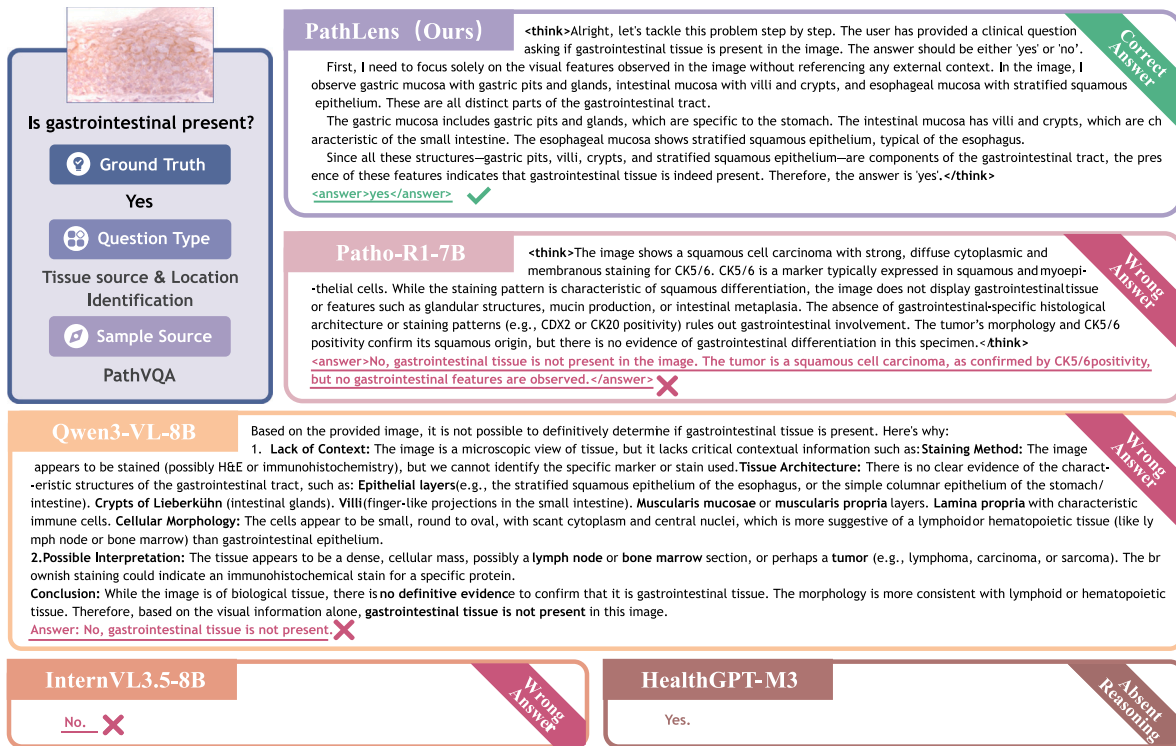


Fig. B.17. The task is to determine whether gastrointestinal tissue is present in the image. PathLens correctly answers yes by jointly recognizing multiple characteristic structures, including gastric pits and glands, intestinal villi and crypts, and stratified squamous epithelium. In contrast, Patho-R1-7B hallucinates an unrelated squamous-cell-carcinoma immunostaining context, Qwen3-VL-8B and InternVL3.5-8B reject the correct answer because of insufficient visual grounding, and HealthGPT-M3 again returns only a terse relevant response. This case shows that PathLens more reliably integrates multiple histological cues into a coherent decision, rather than relying on isolated markers or generic uncertainty patterns.

Table D.10

Hyperparameter configuration for the second-stage ReAL (GSPO) training.

Hyperparameter	Value
Number of training epochs	1
Learning rate	5e-7
KL penalty coefficient β	0.05
Per-device train batch size	32
Number of GPUs	4
Global prompt batch size	128
Number of generations per prompt	4
Maximum input length	1024
Maximum completion length	2048
Sampling temperature	1.0
Warmup ratio	0.01
Precision	bfloat16
DeepSpeed strategy	ZeRO-2

Table E.11

Inference efficiency comparison on PedPathVQA. TPS denotes end-to-end generated-token throughput, computed as the number of generated tokens divided by the total wall-clock inference time, including image encoding, prompt prefill, and autoregressive decoding. Peak memory denotes the maximum GPU memory usage during inference.

Model	TPS $\uparrow$	Peak GPU memory $\downarrow$
Qwen3-VL-8B	20.42	17.08 GB
Patho-R1-7B	22.58	15.92 GB
InternVL3.5-8B	21.48	17.15 GB
LLaVa-Med-7B	19.95	14.60 GB
HealthGPT-M3	5.97	8.52 GB
PathLens	23.02	8.89 GB

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